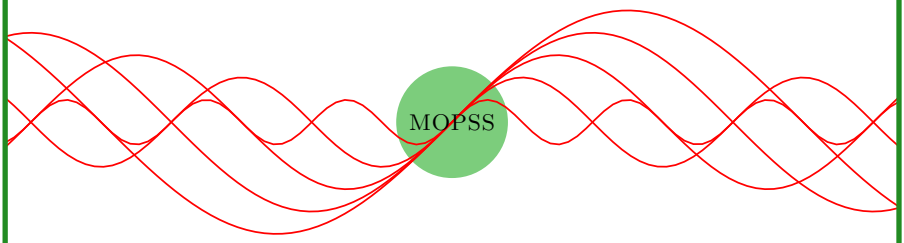
Complex numbers

Mathematics Olympiad

Problem Solving Sessions



Department of Mathematics
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https://jpsaha.github.io/MOTP/MOPSS/

Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (RMO 2015f P2) 2

§1 Using complex numbers

Exercise 1 (RMO 2015f P2)**.** Given that P and Q are points on the sides AB and AC respectively of $\triangle ABC$. The perpendiculars to the sides AB and AC at P and Q respectively meet at D, an interior point of $\triangle ABC$. If M is the midpoint of BC, prove that $PM = QM$ if and only if $\angle BDP = \angle CDQ$.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)